



Modeling the Development of Cellular Exhaustion and Tumor-Immune Stalemate

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Abstract

Cellular exhaustion in various immune cells develops in response to prolonged stimulation and overactivation during chronic infections and in cancer. Marked by an upregulation of inhibitory receptors and diminished effector functions, exhausted immune cells are unable to fully eradicate the antigen responsible for the overexposure. In cancer settings, this results in a relatively small but constant tumor burden known as a localized tumor-immune stalemate. In recent years, studies have elucidated key aspects of the development and progression of cellular exhaustion and have re-addressed previous misconceptions. Biological publications have also provided insight into the functional capabilities of exhausted cells. Complementing these findings, the model presented here serves as a mathematical framework for the establishment of cellular exhaustion and the development of the localized stalemate against a solid tumor. Analysis of this model indicates that this stalemate is stable and can handle small perturbations. Additionally, model analysis also provides insight into potential targets of future immunotherapy efforts.

Keywords Cellular exhaustion · Progenitor exhausted · Terminally exhausted · Stalemate · Immunotherapy

1 Introduction

The adaptive immune response follows a general schematic when defending the body against acute infections. Circulating Naïve T cells encounter cognate antigen when

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displayed by a major histocompatibility complex on an antigen presenting cell (APC) (Sprent 2005). Upon recognition, these Naïve T cells will enter a minimal developmental program to expand their numbers (Kaech and Ahmed 2001). After several replications, these cells will interpret a myriad of environmental signals and differentiate into effector cells, generally CD4+ Helper T cells or CD8+ Cytotoxic T cells (CTLs). After these effector cells migrate to the source of the infection, they will encounter the antigen and respond to this restimulation with proliferation and effector function (Smith-Garvin et al. 2009). The minimal developmental program and the antigen induced proliferation combine to mark the expansion phase of the adaptive immune response. When these T cell populations reach an appropriate size, the immune system can effectively clear the infection; Helper T cells coordinating and sending chemical messengers in the form of cytokines while Cytotoxic T cells kill the targeted infected cells/antigen. Once the infection is cleared, these immune cell populations must be restored back to homeostatic levels to prevent autoimmunity (Seo et al. 2021). A majority of these cells will undergo suppression and subsequent apoptosis, stimulated by chemical cues or with the help of specialized cells, like Regulatory T cells (Wing et al. 2019). This depletion of the immune cell populations defines the contraction phase. T cells that do not commit to programmed cell death will differentiate further into Memory T cells, where they will circulate throughout the body until activated during secondary infections (Akondy et al. 2017). During chronic infections and in cancer, this program shifts. Despite expanded numbers, the antigen may prevail. Failed antigen clearance leads to prolonged stimulation and overactivation of T cells. As a result, the host experiences a phenomenon known as cellular exhaustion. While the evolutionary reasoning for this phenomenon is still debated, it is well documented that cellular exhaustion is a hinderance of the adaptive immune response.

Cellular exhaustion was first observed in murine models during chronic lymphocytic choriomeningitis virus and was later observed in human patients with chronic infections or cancer (Jiang et al. 2021). Exhausted cells differ from normal healthy effector cells, mainly in terms of surface phenotype and functionality. Phenotypes of exhausted cells are characterized by an upregulation of inhibitory receptors, including PD-1, TIM-3, LAG-3, and more (Zhang et al. 2020; Catakovic et al. 2017). While normal healthy effector cells utilize aerobic glycolysis to keep up with energy demands, exhausted cells do not experience the usual metabolic switch, continuing to utilize oxidative phosphorylation and fatty acid oxidation (Sears et al. 2021). Altered surface receptor expression and inefficient energy production cause exhausted cells to provide diminished effector functions. These reduced functions include cytokine production, proliferative ability, cytotoxicity, and more (Wherry et al. 2007; Wherry 2011). Exhausted cells are operational, fighting the chronic infection and or tumor to some degree. However, the weakened abilities of exhausted cells result in a localized stalemate. In tumor settings, the exhausted immune cells keep the tumor at a relatively consistent size as it neither shrinks it nor lets it continue to grow (Wherry and Kurachi 2015). Unfortunately, due to the nature of cancer, tumor cells continue to proliferate to replace the cells that die within the solid tumor. Continued replication during the tumor-immune stalemate provides the tumor the opportunity to evolve and develop new abilities for immune evasion or immunosuppression, allowing it to escape and grow to problematic sizes (Pauken and Wherry 2015).

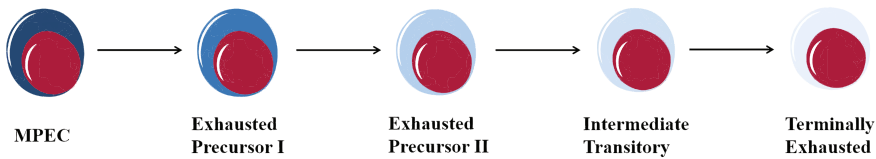


Fig. 1 Differentiation regime of an exhausted CD8+ cell in response to overstimulation, beginning from memory precursor effector cells (MPECs) to terminally exhausted cells

There are several types of cells that become exhausted in cancer settings, yet most research pertains to CD8+ Cytotoxic T cells. In recent years, the complexity of T cell exhaustion has come to light. While it was once thought that exhaustion was a further differentiated state of normal effector T cells, it is now understood that exhausted cells are a distinct lineage coming directly from Naïve T cells (McLane et al. 2019). This exhausted cell lineage is heterogenous, with a series of differentiations that reflect altered phenotypes and hierarchical changes in functional abilities (Pauken and Wherry 2015). This lineage can be seen in Fig 1.

The prolonged exposure to antigen in cancer settings cause the T cell lineage to redirect towards exhaustion. Naïve cells activate and differentiate into memory precursor effector cells (MPECs), similar to normal differentiation in acute infections. During the overexposure and overactivation, subsequent signal transduction in specific biochemical pathways distort the usual differentiation, upregulating inhibitory receptors on the cell surface, upregulating the inhibitory pathways themselves, and thereby promoting exhaustion. This differentiation also includes the metabolic switch previously mentioned. The first stage in the exhausted cell differentiation regime is the Precursor I cell. Continued differentiation leads to Precursor II cells and eventually the Intermediate/Transitory cells. These subsets have moderate inhibitory receptor frequencies on the cell surface. Starting as residents of lymphoid organs, differentiation helps these cells mobilize towards the tumor site. While precursor and transitory cells have little to no cytotoxic abilities, they have some proliferative capacity upon restimulation and can produce moderate levels of key cytokines such as Interleukin 2 (IL-2) (Beltra et al. 2020). As differentiation progresses, cytotoxic ability slightly increases while the remaining effector functions diminish. These subsets of exhausted cells are commonly grouped together for research purposes, a category labeled Progenitor Exhausted cells. Finally, Progenitor Exhausted cells may finish their differentiation program and differentiate into the Terminally Exhausted cells. Terminally Exhausted cells are short-lived and have high expression of inhibitory receptors, mainly PD-1 and TIM-3. These cells have improved cytotoxic abilities compared to Progenitor Exhausted cells while having little to no proliferative capacity or cytokine production (Beltra et al. 2020; Philip and Schietinger 2022; Im et al. 2016; Hudson et al. 2019). The phenotypic landscape of exhausted cells have also been correlated with spatiotemporal dynamics, specifically cell motility in the tumor microenvironment (You et al. 2021). A review of the phenotypic and transcriptional profiles of this exhausted cell lineage have been outlined by Dolina et al. (2021).

To prevent and or reverse this exhausted state of the adaptive immune response, it is paramount that the development, characteristics, and functional abilities of these exhausted cells be understood. Immunotherapy has shown recent advances in the field of targeting cellular exhaustion. Blocking key receptors, like PD-1, which are upregulated on exhausted cells, have been reported to restore effector function in certain settings (Nguyen and Ohashi 2015). Further investigation also suggests that combining this blockade with the blockade of other inhibitory receptors will have synergistic effects in clinical settings (Catakovic et al. 2017; Pardoll 2012). This has been a promising field as of late, providing beneficial results in melanoma, non-small cell lung cancer, renal cell carcinoma, bladder cancer, and more (Pauken and Wherry 2015).

To further improve immunotherapeutic efforts, it is important to gain insight at the cellular or population scale. Analysis of exhausted cell dynamics in cancer settings may help improve current strategies or provide concentrations for future immunotherapy design.

In this study, we make use of mathematical modeling and subsequent model analysis to better understand the dynamics of cellular exhaustion in response to tumors. Evaluating the population dynamics involved may provide useful conclusions that will aid in the development or advancement of immunotherapy strategies.

As dictated by Eftimie et al. (2016) and Baral et al. (2019), mathematical modeling has provided a foundation for new avenues in studying immune cell dynamics across different scales, from single cell to population levels, or across spatiotemporal scales, in different immunological settings. Experimental results help construct and advance these mathematical models to aid in analysis of the complex immune environments. Recently, there has been a lot of activity in math communities for developing and applying such models. In fact, Altan-Bonnet et al. (2020) reiterates the practice of using mathematical models, as well as physical concepts, to study and analyze different features of the immune response.

While mathematical modeling of the immune system has been established, there are no models currently created that accurately represent the phenomenon of cellular exhaustion or how these exhausted cells interact in immune environments. Since cellular exhaustion plays a major role in the immune response to chronic infections and in cancer, and because there is overlap between exhaustion in these two settings (Jiang et al. 2015), previously established models for acute infection can act as a solid basis for model construction. Recent models of interest include those by Kim et al. (2010) and Wilson and Levy (2013) which focus on immune cell localization and cell fate during acute infections, as well as Wodarz (2005) which highlights lytic interactions during infections. As the present study revolves around the immune response to tumors, it is essential to incorporate cancer-based models, including a model of tumor growth under the presence of CD8+ T cells and Natural Killer (NK) cells by Song et al. (2020) and a model that evaluated tumor dynamics under immune regulated conditions developed by Milzman et al. (2021).

The structure of this paper is as follows. In Sect. 2 we develop the mathematical model illustrating the establishment of cellular exhaustion. In Sect. 3 we describe the methodology behind parameter estimation for the developed model. Section 4 observes the resulting stalemate between exhausted immune cells and the tumor and analyzes

Table 1 State variables present in the model of cellular exhaustion in response to tumors

State variable	Description
N	Tumor Volume (mm^3)
K	Residual Healthy CTL Population
B	Progenitor Exhausted T cells
E	Terminally Exhausted T cells
P	Interleukin-2 Cytokine Concentration
Concentrations of each non-tumor related variable are represented in thousands cells or molecules per cubic millimeter	

the dynamics involved. Section 5 recounts key details of tumor-immune dynamics, addresses points of discussion raised from model analysis, and expands the scope of mathematical modeling of cellular exhaustion.

2 A Mathematical Model for T Cell Exhaustion

The model in this study captures the overall dynamics of cell populations to track the progression of exhaustion in CD8+ T cells and how they interact with tumor cells. To capture population dynamics, a compartmental model of Ordinary Differential Equations (ODEs) was developed, highlighting the key interactions between cellular species. This model is provided below along with brief descriptions of the individual terms within each equation. Key underlying assumptions for this model are also addressed.

$$\dot{N}(t) = \alpha N \left(1 - \frac{N}{T_c} \right) - h \left(\frac{\lambda_K N K + \lambda_E N E}{N + K + E + 1} \right), \quad (1a)$$

$$\dot{K}(t) = h P K - \delta_K K, \quad (1b)$$

$$\dot{B}(t) = 2^m s_B N + h P B - r_E B - \delta_B B, \quad (1c)$$

$$\dot{E}(t) = r_E B - \delta_E E, \quad (1d)$$

$$\dot{P}(t) = \rho_K K + \rho_B B - h P (K + B) - \delta_P P. \quad (1e)$$

Each compartment within the model represents a different species interacting in the tumor-immune environment. There are four cellular populations, N , K , B , and E , as well as one key cytokine, P , present throughout the cellular interactions, all of which act as the model's state variables. The non-tumoral species exist as concentrations while the tumor is represented as a volume. The state variables in this model are detailed in Table 1.

The first equation in Eq. (1) indicates the dynamics of tumor growth. Although the tumor-immune interface is complex, involving a myriad of cell types, chemical messengers, environmental features, and more, this model is a simplified representation, highlighting specific interactions that govern the tumor-immune stalemate. In the immediate microenvironment, tumor cells compete against immune cells and local tissue for space, metabolites, nutrients, etc. While the type and location of tumors often

influence its abilities to out-compete these nearby cells, tumor uniqueness was ignored. We assume that, in general, the solid tumor will successfully out-compete these additional cell types, resulting in tumor growth. This growth may be modeled in various ways, an already accepted form being logistic growth. In order to assume a solid tumor growing logistically, it must be assumed that there is a carrying capacity for the tumor burden.

The continual growth of the tumor must reach a capacity brought about by an external factor. In the cancer microenvironment, a common external factor implicated in this capacity includes the angiogenic capabilities of the tumor. To accurately model the tumor associated vasculature that influences the tumor's carrying capacity, incorporation of epithelial cells and key cytokine players like VEGF (Vascular Endothelial Growth Factor) and $\text{TNF}\alpha$ (Tumor Necrosis Factor), to name a few, are required. While modeling such specifications may be more biologically accurate, their negligible interactions with the immune cells directly acting on the tumor in the tumor-immune stalemate suggest that this level of detail is not needed. To compensate, we ignore these additional cells and cytokines and assume the tumor associated vasculature supports tumor growth to a physical predetermined size. We also assume that this vasculature does not allow for possible malignant transformations, confining the model to a single solid tumor interacting with the adaptive immune system. This predetermined size comes from previously published works (see Parameter Estimation Section).

Additionally, through a variety of methodologies, tumor cells have been reported to influence the physical environment to support tumor growth and spread. This includes indirect use of tumor associated macrophages and cancer associated fibroblasts. Reconstructions of the extracellular matrix and environment to create an ideal space for tumor growth consists of several components and additional cell types and chemical messengers. Similar to the previously mentioned tumor associated vasculature, this incorporation into the model will increase complexity while having little to no impact on the immune cell dynamics involved in exhaustion progression or interaction with the tumor. The spatial concepts of tumor growth and physical environment are thus ignored in favor of specifying a physical volume to act as a carrying capacity.

Counteracting tumor growth, cellular death limits the total volume of the tumor burden. Tumor cells die as a result of the tumor-immune interactions. When effector CD8^+ T cells penetrate the tumor, whether they be normal or exhausted, cytotoxic abilities allow the immune cells to lyse tumor cells. These killing terms are derived from cellular saturation-based dynamics, an accepted structure previously used in the models of Eftimie et al. (2011) and Gadhamsetty et al. (2014, 2017). Because Progenitor Exhausted cells have little to no cytotoxic effects on tumors, the only two death terms considered come from interactions with healthy effector cells and Terminally Exhausted cells.

Equation (1b) considers the normal healthy effector CD8^+ Cytotoxic T cell population, K . These cells consist of the immune cells that have already penetrated the tumor prior to the development of exhaustion. When exhaustion initiates, Naïve T cells no longer supply the healthy CD8^+ T cell population, thus there is no influx in this population. However, these cells are effector cells that can replicate via IL-2 induced proliferation in accordance with the law of mass action. Aside from proliferation, this

population may change depending on CD8+ T cell death rates. Death can occur via activation induced cell death, apoptosis, or by other means. These forms of cellular death are grouped together in the second term within the equation.

Equation (1c) marks the changing population of Progenitor Exhausted T cells, B. When exhaustion develops due to prolonged stimulation, Naïve cells enter a minimal developmental program and then differentiate into Progenitor cells. Similar to proliferation, this supply of Progenitor Exhausted cells follows the law of mass action, dependent on the tumor cell population. The first term includes the Naïve cell supply rate and the number of divisions required for completion of the minimal developmental program, resulting in the supply of differentiated Progenitor Exhausted cells. These cells also undergo IL-2 dependent proliferation, encapsulated by the second term. Progenitor cells can also continue to differentiate into the Terminally Exhausted subset, whose population dynamics is represented by the following equation in (1). This differentiation occurs at linear rate as specified in the next section. Lastly, Progenitor cells are subject to cell death much like the normal CD8+ T cell population.

Equation (1d) represents the Terminally Exhausted cells, E, introduced into the system via differentiation from the Progenitor Exhausted cells. Terminally Exhausted cells are short-lived, dying not too long after being fully differentiated. The differentiation that contributes to Terminally Exhausted cell growth and the death of these cells are shown in the first and second terms respectively. These cells have little to no proliferative ability thus no additional terms are needed.

Finally, the last equation accounts for the concentration of IL-2 in the system, P. This cytokine is essential for the proliferation of healthy CD8+ T cells as well as the Progenitor Exhausted cells. IL-2 is produced by effector CD8+ T cells and by Progenitor Exhausted cells in this system. Terminally Exhausted cells produce negligible amounts; thus, it is ignored. Once this cytokine has been produced and secreted, it may leave the system in one of two ways; uptake by the same two immune cell populations as it is used for proliferation, or by natural decay. These two cytokine fates are represented by the third and fourth terms respectively.

3 Parameter Estimation

Model (1) consists of several compartments, each with individual parameters highlighting compartment dynamics. Some of these parameters have accepted values within recent literature while others have been estimated in immunological models. These values, both accepted and estimated generally relate to terms within equations dealing with normal CD8+ effector T cells and the IL-2 cytokine. New parameters dealing with the exhausted cells in Eqs. (1a), (1c), (1d), and (1e) within (1) need to be estimated using new data as there are no previous population-based models dealing with cellular exhaustion. In some instances, estimates for normal healthy effector cells may suffice. The data used for tumor growth and exhausted cell parameter estimation comes from Miller et al. (2019). Estimations have been obtained from a variety of sources, from both murine and human model studies, as well as from studies for infections and tumors. Although this project's parameter estimation pulls from a variety of sources, the estimation process was used to determine a single individual value to be used in

Table 2 Description of parameters within model (1) equations

Parameter	Description
α	Tumor growth rate
T_c	Tumor carrying capacity
λ_K	Tumor-CTL lysis parameter
λ_E	Tumor-Terminally Exhausted lysis parameter
h	Magnifier for physical interactions
δ_K	Death rate of CTLs
δ_B	Death rate of Progenitor Exhausted cells
δ_E	Death rate of Terminally Exhausted cells
δ_P	Decay rate of IL-2
m	Progenitor cell divisions
s_B	Naive cell to Progenitor cell supply rate
r_E	Differentiation Rate
ρ_K	CTL cytokine secretion rate
ρ_B	Progenitor cytokine secretion

simulations, rather than a distribution of values for each parameter. Table 2 provides a description of each parameter. Accompanying this table is the general process behind parameter estimation. Table 3 summarizes the value of each parameter along with the source of value determination.

Tumor Growth

The tumor growth rate, α , was determined via estimation from the data provided by Miller et al. (2019). OVA-expressing B16.F10 melanoma tumor volumes (mm^3) were taken for 18 days in murine models. Due to the duration of these measurements, the tumor data fits the exponential growth seen in the early stages of tumor progression. To fit a logistic curve to this exponential data, a tumor carrying capacity is needed. In previous mathematical immunological models with logistic tumor growth, this parameter takes on a wide range of values. For this model, the carrying capacity, T_c , was chosen to match the average tumor carrying capacity from the scramble control set data from the tumor measurements used in the mathematical model of Milzman et al. (2021). This carrying capacity was set as 4.4725×10^3 . From this data, along with the tumor volume measurements of Miller et al. (2019), parameter estimation using least squared regression provides the tumor growth rate, α , with the value of 0.3 day^{-1} .

Cell Supply

During acute infections, Naïve T cells circulate the lymphatic and circulatory systems. When these cells encounter cognate antigen, there is activation and a subsequent entering of the minimal developmental program before differentiating into full-fledged effector cells, either as CD4+ Helper T cells or CD8+ Cytotoxic T cells. In cancer settings, there is overstimulation and overactivation of the T cell, thus, while the bulk of the previously mentioned T cell fate remains intact, prolonged stimulation results

in the differentiation regime being slightly altered, promoting the development of exhausted cells rather than normal effector cells. For this model, we assume that the supply rate of Progenitor Exhausted cells is equal to the supply rate of effector CD8+ T cells estimated from previous studies. This assumption stems from the notion that in cancer, the Naïve cell supply remains unchanged, and that the overstimulation simply redirects the T cell fate towards exhaustion. Previously, using approximate Naïve T cell turnover data from Mohri et al. (2001), which uses a human HIV-1 infection model, Kim et al. (2010) estimates the value of active Naïve T cell supply to be 0.0012 thousand cells per micro-liter. Following this estimation, the parameter s_B , which in this model represents the supply rate of the Progenitor Exhausted cells, is estimated to be 0.0012 k/day.

Development

Upon activation, T cells undergo a minimal developmental program to expand their population; a program consisting of cellular divisions. It is important to note that this form of expansion is different from cellular proliferation that occurs upon restimulation. While there are arguments suggesting that the minimal developmental program may be time-dependent, the constructed model of this study uses a parameter to define a set number of divisions rather than time. The parameter defining the number of divisions within the minimal developmental program for T cells, m , was chosen to be 7, matching that of Kim et al. (2010), which was previously taken from Kaech and Ahmed (2001) who observed a minimum of 7–10 divisions for T cells in such a program.

Cell Death

While Progenitor Exhausted cells exhibit similar phenotypes to that of Naïve T cells, they are a differentiated cell type. Following the general progression to effector cell, it is assumed that these newly differentiated Progenitor cells would naturally die at greater rates than Naïve cells. According to Kim et al. (2010), who references the data of Mohri et al. (2001) and Boer et al. (2003), it was estimated that Naïve T cells naturally die at a rate 0.03 k/day while fully developed effector CD8+ Cytotoxic T cells die at a rate of 0.4 k/day. This death is generally brought on by apoptosis. The early stages of this process of naturally programmed cell death are correlated with the expression of Annexin V. Miller et al. (2019) data shows that 41% of Progenitor cells express Annexin V while 72% of Terminally Exhausted cells present this biomarker. With this, we assume Progenitor cells die at a similar rate of 0.41 k/day. Additionally, it is estimated that Terminally Exhausted cells die at a rate of 0.72 k/day. It is important to note that this estimate for natural death of Terminally Exhausted cells is much larger than that of the normal healthy effector cells. This appears plausible as Terminally Exhausted cells have a shorter life span than normal effector cells, contributed in part by an increase of inhibitory receptors on their surface, like PD-1, which has also been linked to apoptotic rates.

Differentiation

Spanning 21 days after tumor challenge, Miller et al. (2019) measured the frequency of Progenitor and Terminally Exhausted cells. This data appears to represent the conversion/differentiation of Progenitor cells into the Terminally Exhausted cell subset.

Fitting of this data shows that the Progenitor Exhausted cell frequency decayed exponentially with at a rate of 0.251 k/day while the Terminally Exhausted cells appeared to increase linearly at a rate of 5.42% per day. The differences in these growth patterns may reflect the tetramer labeling that led to the classifications as either Progenitor or Terminally Exhausted cells. Discriminating solely by intermediate and high levels of PD-1 or TIM-3 may be difficult for perfectly distinguishing between Progenitor and Terminally Exhausted cells as other receptor and transcription factor frequencies have been identified for useful subset classification. The different population rates of change may also be influenced by the death rates of each subset. To avoid overcomplication with the tetramer labeling and to account for the different lifespans between Progenitor and Terminally Exhausted cells, the differentiation parameter, r_E , was chosen to be linear, matching that of the Terminally Exhausted cell.

Cytotoxicity

Killing abilities of the tumor-immune interactions are often represented in terms of saturation, capable of handling tumor growth and destruction. Like the tumor lysis parameter of Milzman et al. (2021), which was used to define CD8+ T cell cytotoxicity in murine models challenged with B16 tumors, this model expresses the cytotoxic killing of tumor cells by CD8+ T cells and Exhausted cells via sigmoidal saturation. In the tumor environment, with the development of exhaustion, tumor killing can occur on several fronts. The remaining CD8+ T cells that have already infiltrated the tumor may function as normal effector cells, killing with a lysis parameter λ_K . The model of Milzman et al. (2021) used a variety of values for this parameter. For this present study, the value of 0.111 mm³/k/day, which falls within the range of Milzman et al. (2021), was selected. The cytotoxic ability of CD8+ T cells correlate with the degranulation and secretion of perforin and granzyme B. Polymerization of perforin in the target cell's membrane creates pores and arcs that partially destabilizes the cell as well as allows passage into the cell for Granzyme B (Hoves et al. 2009; Voskoboinik et al. 2015). This cytotoxin is responsible for cleaving and degrading the target from the inside, causing cellular damage and triggering an apoptotic response. In Miller et al. (2019), which also observed B16 tumors in murine models, it was reported that approximately 80% of Terminally Exhausted cells produce this granzyme for cytotoxicity. While Progenitor Exhausted cells have been recorded to produce this cytotoxin, their effect on tumor lysis has been reported to be negligible. Thus, for this model, the additional killing term incorporated into equation (1a) comes from the Terminally Exhausted cells with λ_E estimated to be 0.0888 mm³/k/day, 80% of the normal effector cell counterpart's lysis parameter.

Cytokine Production

According to Gaffen and Liu (2004), immune cells can undergo a base level of IL-2 production upon activation, with the exception of the largest cytokine contributor, the CD4+ Helper T cells, which produce much more. In previously published models, (Kim et al. 2013; Wilson and Levy 2013; Milzman et al. 2021), IL-2 productions rates of immune cells were estimated to be 1.0 k/day. However, from D'Souza and Lefrancois (2004), it has been reported that it is rare for more than 50% of activated immune cells to produce IL-2. If taken together, it can be interpreted that for previous models, 50% of the population of activated cytokine-producing immune cells produced

Table 3 Summary of defined parameter values and their respective sources used for value determination

Parameter	Value	Source
α	0.3 day^{-1}	Estimated
T_c	4472.5 mm^3	Milzman et al. (2021)
λ_K	$0.111 \text{ mm}^3/\text{k/day}$	Milzman et al. (2021)
λ_E	$0.0888 \text{ mm}^3/\text{k/day}$	Estimated
h	10	Fixed
δ_K	0.4 k/day	Kim et al. (2010)
δ_B	0.41 k/day	Estimated
δ_E	0.72 k/day	Estimated
δ_P	5.5 k/day	Kim et al. (2011)
m	7	Kim et al. (2010)
s_B	0.0012 k/day	Estimated
r_E	0.0542 k/day	Estimated
ρ_K	1 k/day	Kim et al. (2010), Wilson and Levy (2013), and Milzman et al. (2021)
ρ_B	0.36 k/day	Estimated

Parameter unit, k, represents thousands

double the IL-2, i.e. 2.0 k/day, to create a total population producing at a rate of 1.0 k/day. Data provided by Miller et al. (2019) indicates that the frequency of IL-2 producing Progenitor cells was 18%, thus the rate of cytokine production for the total population of Progenitor cells, ρ_B , was estimated to be 0.36 k/day, to account for the double IL-2 productivity. Additionally, Terminally Exhausted T cells produce negligible amounts of this cytokine, thus no need for incorporation into this model, nor any parameter estimation. It is also worth noting that extracellular IL-2 decays linearly at a rate of 5.5 k/day, as estimated by Kim et al. (2011). While these cell types have been reported to produce other cytokines needed for the immune response, additional cytokine concentrations are not being considered in the present study.

4 Model Simulation and Analysis

Implementation of model (1) with the parameter values from Table 3 are shown in Fig. 2. This simulation was performed by MATLAB's ode23 function. For this system, we see that the growing tumor provides prolonged T cell stimulation and subsequent development of cellular exhaustion. We also note that variations in initial tumor size may only adjust the time span of system dynamics, not the dynamics themselves. Thus, as the tumor grows, the population of Progenitor Exhausted and Terminally Exhausted cells increase. Any preexisting healthy effector cells eventually leaves the system via cell death. The IL-2 cytokine remains in the system at low levels as it is both produced and utilized by Progenitor Exhausted cells and healthy effector cells. Despite IL-2 dependent proliferation by normal effector cells, the low levels of this cytokine and lack of additional system supply/input are not sufficiently adequate to

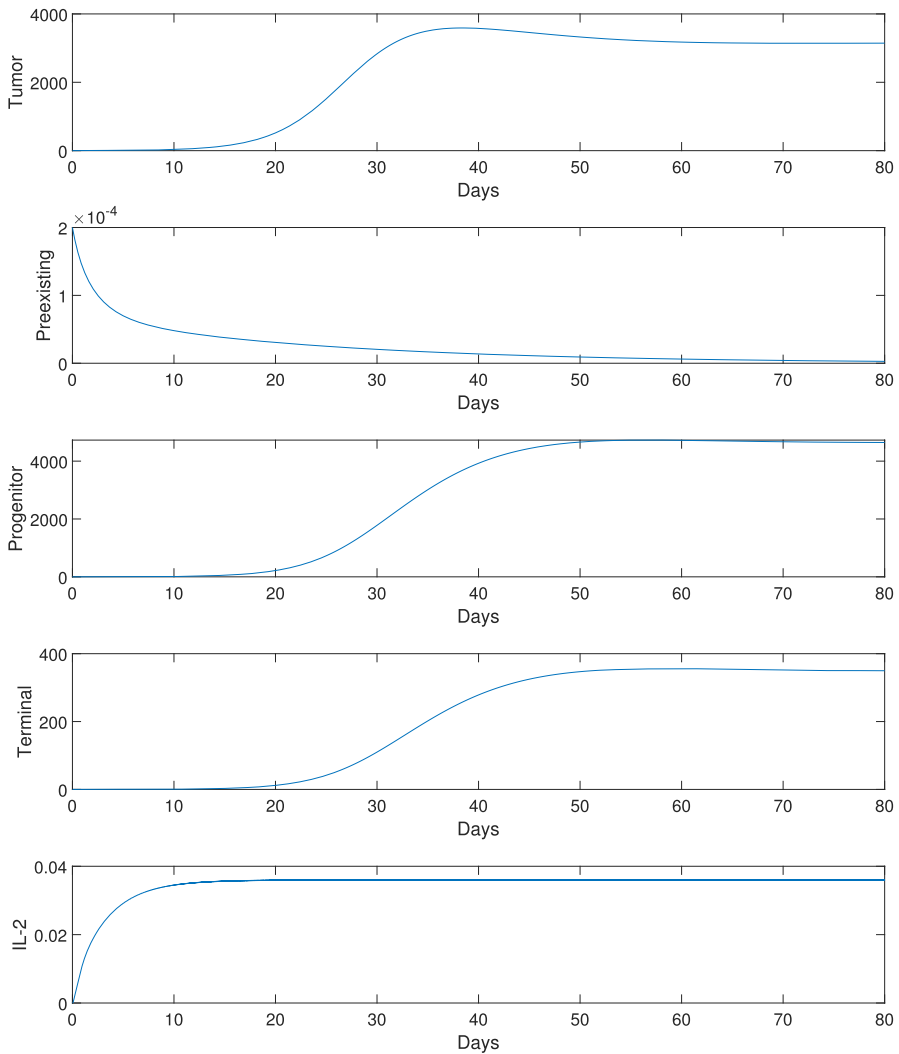


Fig. 2 Model dynamics over 80 days during the development of cellular exhaustion. Shown are: Tumor volumes, normal and healthy effector cells, Progenitor Exhausted cells, Terminally Exhausted cells, and the IL-2 cytokine

surpass the normal rate of CD8+ T cell death, which explains why this population diminishes over time. Eventually, with large enough population sizes of the exhausted cell compartments, a tumor-immune stalemate was reached.

In this system, the tumor volume reaches its peak of 3589.03 mm^3 around day 38. Soon after, in response to the growing exhausted cell populations, the tumor diminishes, eventually reaching the stalemate at an approximate volume of 3150.57 mm^3 . The remaining populations also plateau during this stalemate, forming the coexistent steady state. This steady state is defined in the next section.

4.1 Steady States and Stability

For this non-linear dynamical system, there are several states that can be mathematically defined for which the system compartments reach equilibrium. These defined states can be expressed in terms of the parameters outlined in Table 3. However, each steady state solution contains a set of conditions for parameter values and or the relationship to other parameter values for which the corresponding species is not only defined but also biologically valid. One such condition stems from the assumption that each species within the model must be non-negative. However, due to parameter values contradicting certain conditions required for such states, several steady states are ignored. The remaining biologically significant steady states consist of the trivial solution, a conditional boundary solution, and a solution for which all species coexist.

Biologically, the trivial steady state is irrelevant as it is not only indicative of a cancer-free system, but also of an immune-free system.

For the conditional boundary state, the system is representative of a non-trivial and non-tumor bearing model. Without a tumor present, there is no input of Progenitor Exhausted cells and thus no Terminally Exhausted immune cells. In biological terms, this is appropriate as the absence of a tumor means there is no active antigen responsible for prolonged T cell stimulation and therefore no subsequent development of cellular exhaustion. Although there is no tumor, existing CD8+ cytotoxic lymphocytes are present in the system. As healthy effector CD8+ T cells contribute to IL-2 production for proliferative capacities, this cytokine will also exist in this steady state. The remaining two populations of preexisting CTLs and IL-2 may exist as part of a feed-forward loop in the normal immune surveillance system. If we consider the healthy CTL population at a steady state, we know:

$$\dot{K}(t) = 0 = hPK - \delta_K K. \quad (2)$$

From here, we can see that regardless of the value of the CTL population (K), this compartment can reach equilibrium when:

$$P^* = \frac{\delta_K}{h}. \quad (3)$$

Next, we can use this newly defined value for the cytokine concentration to solve for the effector population by substituting it into the IL-2 compartment, whose dynamics no longer depend on Progenitor Exhausted cells due to their absence.

$$\begin{aligned} \dot{P}(t) = 0 &= \rho_K K - hPK - \delta_P P, \\ K(\rho_K - hP) &= \delta_P P, \\ K &= \frac{\delta_P P}{(\rho_K - hP)}, \\ K^* &= \frac{\delta_P \delta_K}{h(\rho_K - \delta_K)}. \end{aligned} \quad (4)$$

At this boundary steady state, the normal healthy lymphocytes produce IL-2 to be utilized in a feedback loop, with cytokine dependent proliferation maintaining its population size. As these populations are defined by parameter values specified for the model, calculations provided explicit population sizes. In this boundary steady state, the healthy normal effector cell population resides at a value of 0.367 k/mm^3 while the IL-2 cytokine concentration remains at 0.04 k/mm^3 .

The most interesting of the steady states is the coexistence state. To recapitulate the system dynamics leading to the coexistent steady state, the initial exponential growth of the tumor is accompanied by the growth of the Progenitor Exhausted cell population. Additionally, as the Progenitor Exhausted cell population grows, continued differentiation causes a subsequent increase in the Terminally Exhausted population. Any preexisting healthy CD8+ effector cells will either undergo cytokine-dependent proliferation or die. According to the biological implications of cytokine production and sequestration, the Interleukin-2 cytokine will remain at extremely low levels within the system. Simply put, the cytokines produced into the system are almost immediately removed via natural decay or via cellular uptake during the cytokine dependent proliferation. As the dynamics of the normal healthy effector CT8+ CTLs depend on the IL-2 concentration, it can be expected that this population also remains extremely low. When considering the other parameter values of effector cell dynamics (physical interactions magnifier and cellular death rate), the population of CTLs tends toward 0, without fully being eliminated. As this population shrinks towards 0, the IL-2 concentration can be reevaluated, giving:

$$\dot{P}(t) = \rho_B B - h P B - \delta_P P. \quad (5)$$

During a potential steady state, this concentration will not change, thus we have:

$$\begin{aligned} \dot{P}(t) = 0 &= \rho_B B - h P B - \delta_P P, \\ \text{i.e., } \delta_P P &= (\rho_B - h P) B. \end{aligned} \quad (6)$$

In this system, it is assumed that each state variable coexists, thereby meaning each population value is positive. As each parameter value is also positive, we see that in order for this expression to be true, a new condition is imposed.

$$\begin{aligned} (\rho_B - h P) &> 0, \\ P &< \frac{\rho_B}{h}. \end{aligned} \quad (7)$$

Thus, during the steady state for which all cellular and cytokine species exist, the population of CD8+ effector cells tends toward 0 and the IL-2 cytokine approaches the value defined by parameters in the final line of (7); a value of 0.036 k/mm^3 (after substituting parameter values). Despite the near elimination of the normal effector cell population, the increase in Terminally Exhausted cells compensates for tumor lysis.

As a result of the increased exhausted cell populations, the tumor never reaches its carrying capacity. Instead, the Terminally Exhausted cell population passes a threshold and can begin to shrink the tumor. However, as a consequence of cellular exhaustion,

these immune cells do not possess the capabilities to fully eliminate the tumor, resulting in a stalemate. This tumor stalemate represents the coexistent steady state. In this model, the tumor immune stalemate is reached around day 60. As previously discussed, this stalemate occurs when the tumor shrinks towards a volume of 3150.57 mm^3 . At this stalemate, with normal healthy effector cells and the IL-2 cytokine population values defined, steady state population values of the remaining cellular species may be determined. Starting with Progenitor Exhausted cells:

$$\begin{aligned}\dot{B}(t) &= 2^m s_B N + h P B - r_E B - \delta_B B = 0. \\ B &= \frac{2^m s_B N}{r_E + \delta_B - \rho_B}.\end{aligned}\quad (8)$$

Here, the Progenitor Exhausted cell population at the coexistent steady state is a function of the tumor, N . As it is assumed that the tumor exists, there by making N positive, (8) implies:

$$r_E + \delta_B - \rho_B > 0. \quad (9)$$

Based on parameter values specified in Table 3, equation (9) is satisfied. Additionally, as the Terminally Exhausted cells are directly related to the Progenitor Exhausted cells, they too can be written as a function of N , under the same condition previously described.

$$\begin{aligned}\dot{E}(t) &= r_E B - \delta_E E = 0. \\ E &= \frac{r_E B}{\delta_E} = \frac{2^m s_B r_E N}{\delta_E (r_E + \delta_B - \rho_B)}.\end{aligned}\quad (10)$$

Finally,

$$\dot{N}(t) = \alpha N \left(1 - \frac{N}{T_c}\right) - h \left(\frac{\lambda_k N K + \lambda_E N E}{N + K + E + 1}\right). \quad (11)$$

As the normal healthy CTL population tends to 0 and the Terminally Exhausted group is a function of N , the resulting equation can be expressed as a quadratic function whose coefficients consist of parameter values.

$$0 = \left[\frac{-\alpha}{T_c} - \frac{2^m \alpha s_B r_E}{T_c \delta_E (r_E + \delta_B - \rho_B)} \right] N^2 + \left[\alpha - \frac{\alpha}{T_c} + \frac{2^m s_B r_E (\alpha - h \lambda_E)}{\delta_E (r_E + \delta_B - \rho_B)} \right] N + \alpha. \quad (12)$$

Solving (12) for N , using MATLAB's "fsolve" function, provides the tumor size at the coexistent steady state of the system, a value of 3150.57 mm^3 . With this value determined, it is then possible to explicitly define the two exhausted cell populations, which are functions of the tumor. The steady state population sizes for each non-tumor species within this model, as functions of the tumor and parameters, are summarized

Table 4 Numerical values for population sizes for the tumor, immune cells, and IL-2 cytokine at the stalemate

State variable	Population size
N	3150.57 mm ³
K	~0
B	4644.22 k/mm ³
E	349.61 k/mm ³
P	0.036 k/mm ³

below. Additionally, the population sizes at this steady state were also calculated and provided in Table 4.

$$K^* \rightarrow 0, \quad (13a)$$

$$B^* = \frac{2^m s_B N}{r_E + \delta_B - \rho_B}, \quad (13b)$$

$$E^* = \frac{2^m s_B r_E N}{\delta_E (r_E + \delta_B - \rho_B)}, \quad (13c)$$

$$P^* \rightarrow \frac{\rho_B}{h}. \quad (13d)$$

To evaluate the stability of this coexistent steady state, the corresponding Jacobian was analyzed. The Jacobian matrix was calculated analytically and analyzed computationally. Below is the Jacobian Matrix for the nonlinear system of ordinary differential equations representing cellular exhaustion in Cytotoxic T cells in response to a growing solid tumor. For simplicity, all parameters and state variables remain arbitrary in this 5x5 matrix.

$$\begin{vmatrix} X_1 & X_2 & 0 & X_3 & 0 \\ 0 & hP - \delta_K & 0 & 0 & hK \\ 2^m s_B & 0 & hP - r_E - \delta_B & 0 & hB \\ 0 & 0 & r_E & -\delta_E & 0 \\ 0 & \rho_K - hP & \rho_B - hP & 0 & h(K + B) - \delta_P \end{vmatrix}$$

For organizational purposes, three elements of the Jacobian Matrix are represented as variables. These variables are defined as:

$$\begin{aligned} X_1 &= \alpha \left(1 - \frac{2N}{T_c} \right) = h \frac{(\lambda_K K + \lambda_E E)(N + K + E + 1) - N(\lambda_K K + \lambda_E E)}{(N + K + E + 1)^2}, \\ X_2 &= -hN \frac{(\lambda_K)(N + K + E + 1) - (\lambda_K K + \lambda_E E)}{(N + K + E + 1)^2}, \\ X_3 &= -hN \frac{(\lambda_E)(N + K + E + 1) - (\lambda_K K + \lambda_E E)}{(N + K + E + 1)^2}. \end{aligned} \quad (14)$$

The eigenvalues of the Jacobian matrix at the coexistent steady state solution were calculated using MATLAB's "eig" function. These eigenvalues are as follows:

$$\begin{aligned} & -0.11002 + 0.09720i, \\ & -0.11002 - 0.09720i, \\ & -0.73570 + 0i, \\ & -46447.72809 + 0i, \\ & -0.04 + 0i \end{aligned} \quad (15)$$

Since the real parts of the eigenvalues of the Jacobian matrix are negative, this particular steady state is asymptotically stable. This suggests that with small perturbations to the system, tumor growth or tumor control would resume (depending on the perturbation), driving the tumor-immune system back towards the localized stalemate. Due to the presence of complex conjugates within these eigenvalues, the system experiences an oscillatory behavior surrounding this stalemate. These oscillations are the result of a nonlinear dynamical system. Oscillations are also influenced by physical interactions between species. This impact is mentioned in further detail in the Discussion Section. Although complex conjugates were present, they did not appear in the dominant eigenvalue, thus there is no spiral focus for this coexistent steady state.

Together, this model highlights the development of cellular exhaustion in immune cells in response to solid tumors. Following the development and progression of exhaustion in this system, the immune cells reach a localized stalemate with the tumor. This stalemate is a stable fixed point and is capable of handling slight perturbations. The lack of bifurcations within this model suggests that there are no parameters capable of destabilizing the localized stalemate after having undergone slight changes. Targeting certain parameters as a form of external influence or therapeutic effort will require differences large enough to overcome the stability of this stalemate.

4.2 Sensitivity Analysis

As the parameters within this model represent key features of immune cells in the tumor-immune environment, their respective values help dictate overall dynamics. In recent years, there has been a large push for immunotherapeutic efforts to target such features of these exhausted immune cells. The goal of these efforts are to improve immune cell capabilities so as to effectively reduce and or eliminate the tumor. Naturally, these features of the immune cells operate differently, biologically or chemically, thus their impact on tumor dynamics vary. To improve future immunotherapy regimes, it is important to understand which aspects of exhausted immune cells contribute the most, directly or indirectly, to tumor control, thereby effectively prioritizing immunological features to target.

By performing a sensitivity analysis on selected parameters within this project's model, the impact of each parameter on the tumor may be evaluated. For such analysis, the parameters revolving around the exhausted cell compartments have been selected. These parameters include the Terminally Exhausted cell lysis parameter, λ_E ,

Table 5 Sensitivity Coefficients for model parameters involving exhausted cells

Parameter	Sensitivity coefficient
λ_E	1.22×10^3
r_E	4.98×10^2
δ_B	6.74×10^3
δ_E	1.20×10^3
ρ_B	5.52×10^3

the exhausted cell differentiation rate, r_E , the Progenitor and Terminally Exhausted cell death rates, δ_B and δ_E respectively, and finally the cytokine production rate of Progenitor cells, ρ_B . It is important to note that these exhausted cell parameters were estimated (see Parameter Estimation Section). As a result, this sensitivity analysis reflects how this model system reacts to perturbations away from the current estimated value. As the study into cellular exhaustion continues, new data may be incorporated to further aid in parameter estimation, which will require subsequent analysis to address potential shifts in system sensitivities.

Independently, tumor volumes at day 100 were examined following perturbations of the selected parameters. The difference between the newly perturbed tumor volume and the original tumor size determined by the model were calculated. As each parameter represents a different aspect of the immune cell, cross comparisons are difficult. To eliminate these difficulties, normalizing the change in tumoral dynamics by the magnitude of the perturbation provides a single non-dimensional measurement, the sensitivity coefficient:

$$SeCo = \frac{N - N_p}{|Perturbation|}, \quad (16)$$

Here, N_p denotes the tumor volume after the individual parameter has been perturbed while N specifies the tumor volume sprouting from the original parameter values set in Table 3.

Depending on the parameter and the direction of the parameter perturbation, the resulting tumoral dynamics may promote tumor growth or tumor reduction. For example, a slight increase in the Terminally Exhausted cell lysis parameter results in a smaller tumor volume while a slight increase in the exhausted cell death rate will result in tumor growth. To keep comparisons consistent, the sensitivity coefficient was calculated using a parameter perturbation of 10% that corresponds to tumor regression. The sensitivity coefficients are provided in Table 5.

When considering a 10% perturbation, the parameter that had the largest impact on tumor shrinkage was the Progenitor Exhausted cell death rate, δ_B . The next parameter for which the cellular exhaustion model was most sensitive to was the IL-2 cytokine production rate by Progenitor Exhausted cells, ρ_B . These two parameters are large players in the maintenance of the Progenitor Exhausted cell population. Taken together, this suggests that this compartment is crucial for tumor-immune interactions. This idea is further supported by several immunotherapy studies regarding PD-1 blockade.

This inhibitory receptor has been linked to both cytokine production and cell death. Blocking this inhibitory receptor partially restored effector function, increasing IL-2 production and decreasing the rate of cell death. As a result of this blockade, the Progenitor cell populations in these studies grew, providing beneficial results in some patients (see Im et al. 2016; Nguyen and Ohashi 2015; Pardoll 2012).

Furthermore, the third most influential exhausted cell parameter in this model was the Terminally Exhausted cell-tumor lysis parameter. This parameter represents the direct killing of tumor cells that lead to diminished tumor volumes. Although this parameter is directly responsible for the intended shrinkage of the tumor volume, its sensitivity coefficient falling short of the parameters that maintain the Progenitor cell populations leads to an interesting conclusion. Since the Progenitor Exhausted cell population differentiates into the Terminally Exhausted population, which kills the tumor cells, it appears that the number of cells outweighs the cytotoxic capabilities of the individual cell. Thus, to improve tumor killing abilities of the exhausted immune response, the priority should be to increase the number of cytotoxic T cells, mainly the Progenitor Exhausted cells as they can contribute to the maintenance of its own population as well as to the growth of the Terminally Exhausted cells. Although the results of this analysis suggests that increasing population size will help combat the tumor, this concept is a double edged sword. This idea is revisited in the Discussion Section.

As briefly discussed, the perturbation of each parameter can have important implications for the overall model. Depending on the parameter, an increase or decrease in value may result in either tumor growth or tumor regression. Additionally, the size of the parameter perturbation may also influence the degree of such tumor growth/regression. To evaluate the intensity of each perturbation and its overall impact on tumoral dynamics, sensitivity coefficients were also calculated for perturbations of 10%, 20%, and 30% in both directions. Due to the overall stability of this cellular exhaustion model, resulting changes in tumor volume may not fluctuate too greatly from one perturbation to the next. These measurements are provided below in Table 6.

If the tumor shrunk after the parameter perturbation, the sensitivity coefficient would be positive. If the tumor grew as a result of the direction of the perturbation, the coefficient would be negative. Overall, the magnitude refers to the size of the impact while the sign refers to the direction of tumoral dynamics. The ranges of these sensitivity coefficients for the exhausted cell parameters are fairly consistent. Slight variations in the orders of magnitude suggest that seeking simple changes in a given parameter adds a relatively small benefit to potential outcomes, further supporting the stability of the stalemate. Large enough perturbations beyond an unknown threshold may be able to disengage from this stalemate, however this concept lies beyond the scope of the current project.

5 Discussion

Cellular exhaustion is a phenomenon observed in immune cells when fighting off cancer, resulting in a hindrance of their functional capabilities. Consequently, diminished immune-related abilities result in a localized stalemate between the immune cells and

Table 6 Spectrum of sensitivity coefficients corresponding to exhausted cell parameter perturbations ranging from -30 to $+30\%$

Param.	-30%	-20%	-10%	10%	20%	30%
λ_E	-3.67×10^3	-2.44×10^3	-1.22×10^3	1.22×10^3	2.43×10^3	3.64×10^3
r_E	-1.90×10^3	-1.19×10^3	-5.58×10^2	4.98×10^2	9.45×10^2	1.35×10^3
δ_B	2.92×10^4	2.92×10^4	6.74×10^3	-3.20×10^3	-5.07×10^3	-6.30×10^3
δ_E	4.47×10^3	2.67×10^3	1.20×10^3	-1.01×10^3	-1.86×10^3	-2.60×10^3
ρ_B	-5.90×10^3	-4.69×10^3	-2.90×10^3	5.52×10^3	2.92×10^4	2.92×10^4

Negative coefficients indicate tumor growth while positive values represent tumor regression

the tumor. Targeting cellular exhaustion as a form of immunotherapy has been a relatively new, but growing field, with promising results. To date, there are no accurate mathematical representations of exhaustion development and progression. The model proposed in this work serves as a preliminary mathematical framework highlighting the major components of this phenomenon during the tumor-immune interactions. The mathematical modeling and subsequent analysis of this system provides a complementary approach to better understand this biological process. Additionally, this framework serves as a solid foundation for continued work in modeling cellular exhaustion. Evaluation of the current model and its limits, as well as model expansion into more accurate depictions of the biological systems, will provide insight into the usefulness of targeting this phenomenon for immunotherapy purposes.

In this model, the magnifying parameter that amplifies the effects of physical interactions plays an important role in the dynamics. For example, increasing this parameter value means that the IL-2 dependent proliferation of Progenitor Exhausted cells would result in increased progeny, thus faster immune cell population growth. However, this increased magnifier also results in increased reactions between the tumor and the cytotoxic abilities of the normal effector cells and the Terminally Exhausted cells. As a result, the tumor shrinks, although only temporary. Tumor shrinkage then results in decreased T cell activation as there is less stimulation of the T cell receptor (TCR) by the tumor, leading to decreased inputs of newly activated Progenitor cells, allowing the tumor growth to resume. Overall, the combined results of an increased magnifier for physical interactions displayed an oscillatory behavior of tumor growth. Over time, as the system appears to stabilize, the oscillations caused by this parameter decrease in amplitude, as depicted below in Fig. 3.

Sensitivity analysis of this model established a ranking of the strengths of the different aspects of cellular exhaustion, mainly their functional abilities. A major takeaway from this analysis was the importance of the Progenitor cell population size. The larger this population, the larger the beneficial impact on the tumor. However, prioritizing this aspect and targeting the improvement of this population size may prove troublesome. Increasing the Progenitor cell population requires strained resource allocation. The physical space and energy demands needed for sustained increases in this immune cell population will need to be considered and monitored in order to avoid unintended consequences. There is also the issue of autoimmunity. There are theories suggesting that exhaustion develops in order to avoid autoimmunity as a result of the high sustained normal effector cell presence. It is unclear whether increased exhausted immune cells would result in the same fate, given their diminished functionality. Further research will be needed.

Additionally, the larger the population of Progenitor cells, the larger the probability for another phenomenon known as cytokine release syndrome, or cytokine storm. Not attributed to this model, it is well understood that these immune cells produce other cytokines outside of IL-2, one of which being IL-10 whose increased concentrations have been reported to have both anti-tumor and exhaustion promoting effects (Hanna et al. 2021). In general, a large influx of extracellular cytokines may lead to detrimental impacts. Excess inflammatory signals may exaggerate the many functions of different cells. This syndrome varies in degree, ranging from mild symptoms like fever, fatigue, and more, to more severe forms, such as organ failure. Due to the diminished produc-

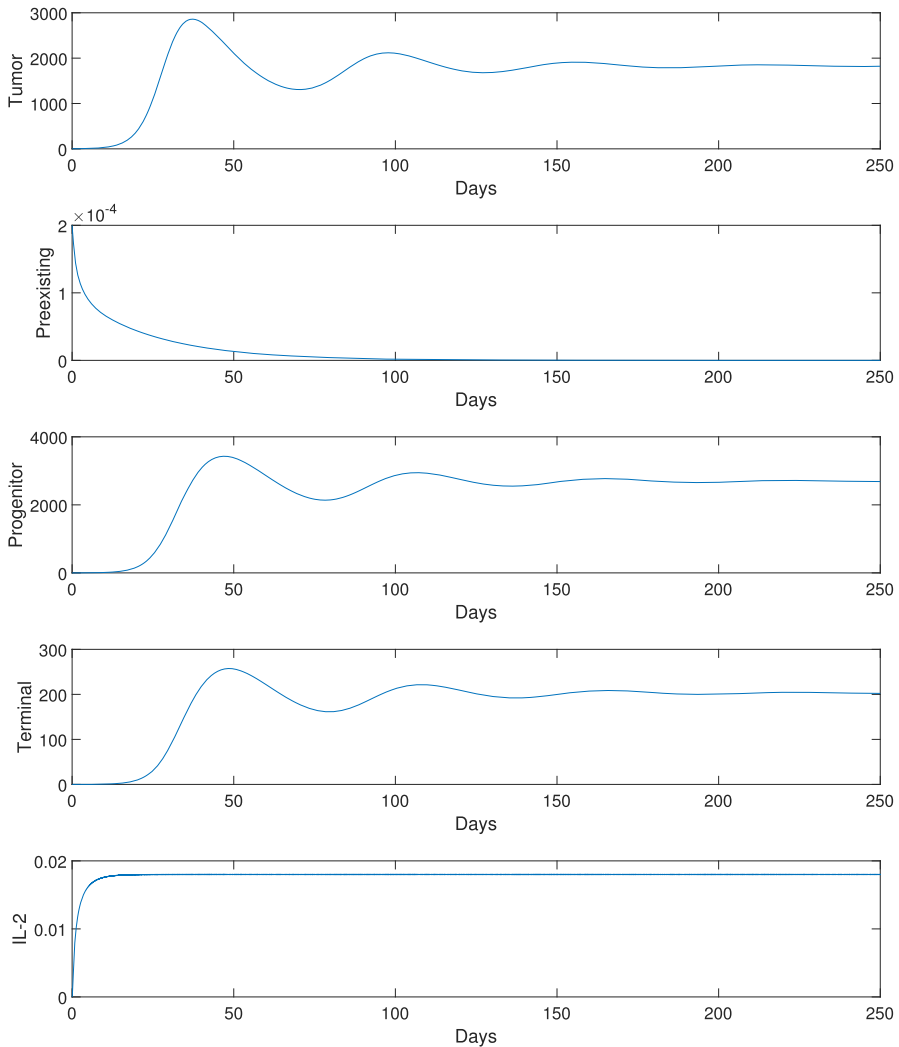


Fig. 3 Oscillations of each population in response to a larger magnification for physical interactions ($h = 20$). Amplitudes of oscillations vary depending on the population

tion of cytokines by exhausted cells, the likelihood of developing cytokine release syndrome while under exhaustion remains unknown. While the sensitivity analysis provides validation and or useful insight into potential targets for immunotherapy, further testings should continue to avoid the risk of the cytokine storm.

This model serves as a preliminary mathematical framework for analyzing and understanding the development, progression, and functional abilities of exhausted cells in response to solid tumors. This foundational model may evolve as the research into cellular exhaustion continues. Future improvements may incorporate additional cell types, like CD4⁺ T cells and Regulatory T cells, as well as additional cytokine

tracking, to represent a more holistic view of the tumor-immune interactions. Parameter estimation will also be more accurate as the research into exhaustion in humans continues. Additional models may be derived from this current model, emphasizing certain aspects of exhaustion such as cell lineage initiation and differentiation stemming from the TOX transcription factor, as it has been identified as a major culprit in the development of cellular exhaustion (Alfei et al. 2019; Khan et al. 2019; Liang et al. 2021). Thus, modeling how exhaustion starts at the transcriptional level may provide useful insight for methods of prevention. In all, this framework for cellular exhaustion in tumor bearing models acts as a foundation for which future model construction and analysis will help identify how targeting exhaustion may be a useful endeavor for immunotherapeutic efforts.

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Declarations

Ethical Approval Not applicable.

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